DATA MANAGEMENT CARPENTRY · EPISODE 3

Naming Files Well

Designing machine-friendly, human-readable filenames that support sorting, batch processing, and long-term reproducibility.

EPISODE SCOPE

Questions & Objectives

Core Questions

What makes a good filename?

How much info belongs in a filename?

Why are naming conventions critical?

How to support batch automation?

Learning Objectives

Identify issues with poorly named files.

Apply sorting and batch processing rules.

Rename individual files using GUI tools.

Bulk rename multi-file sequences reliably.

CURRENT CHALLENGE

From Folder Names to File Names

INHERITED FILE LIST

```
Data!@#$.csv
Untitled.csv
copy of DSC01023.png
03_04_26_results.csv
results-new.csv
data_final.xlsx
data_final_v2.xlsx
data_final_v3_LAST_ONE.xlsx
```

The "Latest Data" Trap

"Can you update the analysis using the latest data?"

> Be wary of relying on date modified timestamps — the last modified file is not necessarily the last valid file!

A filename should reveal what a file contains without requiring you to open it.

DISCUSSION 1

What Problems Can You Identify?

Working in pairs, look through the inherited dataset files:

- Can you tell what the file contains?
- Can you identify the newest version?
- Are dates presented consistently?
- Would the names be easy to process automatically?
- Are there confusing abbreviations or symbols?

COMMON PITFALLS

Filename issues

Meaningless & Vague
data.csv, Untitled.csv, notes.txt

Special Characters
image#1.png, Data!@#$.csv, spaces

Inconsistent Dates
23-Nov2024.csv vs 2024-15-07.txt vs 03_04_26.csv

Ambiguous Version Stacking
data1bfinalfinal2reallyfinalusethis.txt

Duplicate Information
24Nov_image01_241117.dat

Overly long names
Data1256-johnsedit_unreviewed_collectedinaberystwyth_ohwaitthereismore_t.

CORE MOTIVATION

Why Filenames Matter

Five Key Questions Answered

1. **What** is this file?
2. **When** was it created?
3. **Which sample** or dataset does it belong to?
4. **Which version** is active?
5. **Can it group** with related files?



Dual Readability

A filename must serve both **humans** and **computers** across Windows, macOS, and Linux.

Remember: Future-you is your most important collaborator.

EXERCISE

Challenge: Rename an Example File

TARGET FILE

Find: Data!@#$.csv

Rename to descriptive format:

2026-04-03_incidence_data_v01.csv

GUI Shortcuts

- **Windows:** Select file > Press F2 (or right-click Rename)
- **macOS:** Select file > Press Return (or click filename)

Keyboard shortcuts are significantly faster once comfortable.

CORE STANDARDS

Characteristics of Good Filenames

Meaningful

Describes content without opening.

Consistent

Same pattern used everywhere.

Machine-Friendly

No spaces or special symbols.

Sortable

Chronological ISO dates & padded zeros.

Clear

Clean delimiters for parsing.

PORTABILITY

Machine-Friendly Filenames

Safe Characters

Stick to letters, numbers, hyphens, and underscores.

```
my_data.csv / results_and_notes.csv
```

Avoid:

```
spaces and \ / : * ? " < > | & $ #
```

⚠

The OS Case Sensitivity Trap

Linux and macOS are case-sensitive. Windows is case-insensitive.

Copying data.csv and Data.csv from Linux to Windows can overwrite and permanently delete your research data!

LOGICAL ORDERING

Chronological Sorting (ISO 8601)

Poor: Day-First Sorts by Day Number

```
01-02-2025_recording.csv
01-03-2025_recording.csv
01-05-2023_recording.csv
03-04-2026_recording.csv
```

Good: YYYY-MM-DD Aligns Chronology

```
2023-05-01_recording.csv
2025-02-01_recording.csv
2025-03-01_recording.csv
2026-04-03_recording.csv
```

DISCUSSION 2

Ordering Your Project Files

Consider the metadata you collect for one of your own research projects:

👉 **What metadata would you put first to group your files logically?**

Options: Date (`YYYY-MM-DD`), Experiment ID, Sample ID, Subject, or Location.

DESIGN BOUNDS

The Documentation Trap

Overly Verbose Filename

```
2026-06-01_12-43-16_sampling_trip_siteA_temperature_sensor_03_processed_final_v2.csv
```

Risk: Path length overflow and unreadable directory listings.

The Balanced Alternative

```
2026-06-01_siteA_sample03_v01.csv
```

Golden Rule: Include enough to identify the file, but don't turn the filename into documentation. Use READMEs or data dictionaries for details.

AUTOMATION

Leading Zeros in Sequences

Without Padding: Alphabetical Scramble

```
image_1.png
image_10.png
image_2.png
image_8.png
image_80.png
```

With Padded Zeros: Preserved Order

```
image_01.png
image_02.png
image_08.png
image_10.png
image_80.png
```

Rule: If total files < 100, use 2 digits (01–99). If 100–999, use 3 digits (001–999).

PRODUCTIVITY

Bulk Renaming Multiple Files



Windows

- Microsoft PowerToys (PowerRename)
- Bulk Rename Utility / Advanced Renamer



macOS

- Built-in Finder rename tool
(Select multiple files > Right click > Rename)



Linux

- Built-in File Manager tools
- Thunar Bulk Renamer / GPRename

EXERCISE & SOLUTION

Challenge: Bulk Renaming Image Sequences

TASK: Replace IMG with siteA_image_

```
IMG_001.png
IMG_002.png
IMG_003.png
IMG_004.png
IMG_005.png
```

SOLUTION: Standardised Sequence

```
siteA_image_001.png
siteA_image_002.png
siteA_image_003.png
siteA_image_004.png
siteA_image_005.png
```

SAFETY CHALLENGE & SOLUTION

Mitigating Bulk Rename Risks

What Could Go Wrong?

- Accidental overwriting of files with identical names
- Inadvertently stripping critical file extensions
- Irreversible modification to hundreds of records

Risk Mitigation Checklist

- **Always check the tool's preview** before applying.
- **Test on a copy** of a small sample first.
- **Maintain untouched backups** of the raw data.

STRUCTURAL HARMONY

Using Hierarchies Effectively

Redundant Folder & Filename Duplication

```
project/case_studies/scotland/
  └── scotland_2026-06-01_image_01.png

project/case_studies/england/
  └── england_2026-07-02_image_01.png
```

Clean Separation: Context in Folder

```
project/case_studies/scotland/
  └── 2026-06-01_image_01.png

project/case_studies/england/
  └── 2026-07-01_image_01.png
```

Good folder structures and filenames work together — eliminate redundant repetition.

REVISIONS

Explicit File Versioning

The "Final" Trap

```
final.docx final_final.docx

FINAL_ACTUALLY_FINAL.docx
```

Avoid subjective labels like "latest", "new", or "final".

Explicit Version Identifiers

```
report_v01.docx
report_v02.docx
report_v03.docx
(or semantic: report_v1.2.0.docx)
```

Clear sequence that never becomes inaccurate when edited.

Alternative tools: Git, GitHub, Microsoft Office Version History, and OneDrive automatically track changes without file proliferation.

PRACTICAL EXERCISE & SOLUTION

Challenge: Rename the Inherited Dataset

The Assignment

Review the inherited project files and apply our principles:

- Meaningful descriptions
- ISO dates (`YYYY-MM-DD`)
- Explicit version numbers (`_v01`)
- Remove special characters and spaces

PRACTICAL EXERCISE & SOLUTION

Challenge: Rename the Inherited Dataset

A Reasonable Outcome

Before: Data!@#$.csv

After: incidence_data_v01.csv

Before: data_final_v3_LAST_ONE.xlsx

After: project_results_v03.xlsx

SUMMARY

Episode 3 Key Points

Dual Audience: Good filenames help both humans and automated computer scripts.

Consistency Over Perfection: Pick a standard convention and stick to it universally.

Sortability: Use ISO dates (YYYY-MM-DD) and pad numbers with leading zeros.

Explicit Versioning: Replace "final" or "latest" with sequential numbers (_v01).

The Invariant: A good filename identifies content immediately without needing to open the file.

LOOKING AHEAD

Next: Episode 4

We have organised the folders and cleaned the filenames. But how was the data collected, and what do the variables mean? That information belongs in documentation.

Documentation and Metadata